

Security incident report

Section 1: Identify the network protocol involved in the incident

The hypertext transfer protocol is being manipulated in this incident. Since the issue was with accessing the web server for yummyrecipesforme.com, we know that requests to web servers for web pages involve http traffic. Also, when we ran tcpdump and accessed the yummyrecipesforme.com website the corresponding tcpdump log file showed the usage of the http protocol when contacting the website. The malicious file is observed being transported to the users' computers using the HTTP protocol at the application layer.

Section 2: Document the incident

A former employee executed a simple brute force attack on the web host. After they obtained the login credentials, they were able to access the admin panel and change the website's source code. They embedded a javascript function in the source code that prompted visitors to download and run a file upon visiting the website. After embedding the malware, the hacker changed the password to the administrative account. When customers download the file, they are redirected to a fake version of the website that contains the malware. Multiple customers emailed yummyrecipesforme's helpdesk. They complained that the company's website had prompted them to download a file to access free recipes. The customers claimed that, after running the file, the address of the website changed and their personal computers began running more slowly. In response to this incident, the website owner tries to log in to the admin panel but is unable to, so they reach out to the website hosting provider.

Section 3: Recommend one remediation for brute force attacks

I highly recommend enforcing a 2-factor authentication for better security measures going forward because it helps identify the owner of the account as the only one accessing it. One security measure the team plans to implement to protect against brute force attacks is to disallow previous passwords from

being used. Since the vulnerability that led to this attack was the attacker's ability to use a default password to log in, it's important that we prevent any old passwords such as default passwords from being used to reset the password. Another supportive measure is to require more frequent password updates, so in case any unauthorized person becomes aware of the password, they are less likely to be able to use that password if the password is updated sooner than later.